

Acknowledgments

A **very** special thank you to:

Illinois Institute of Technology

Reed Kroloff, Dean of College of Architecture

Vedran Mimica, Director of M.S. Architecture Program

Cynthia Torres, Co-Terminal Graduate Advisor

Jonathan Miller, Associate Professor

Joel Putnam, Adjust Professor

Maria Villalobos Hernandez, Studio Professor

I would not be here without any of you supporting my studies.

Nevertheless, thank you to my parents, family, and friends who have motivated and supported me throughout my years of study, you have been my anchors throughout my academic journey.

Preface

Over the past century, Earth has been faced with the issues stemmed from rapid urbanization. Currently, about half of the global population lives in urban areas. It is estimated that by 2050, 65% of the global population will reside in these urban areas. A new challenge has is emerging amid this urban sprawl: the lack of green spaces. These green oases play a significant role on public health and on the environment.

The Green Prescription for Urban Environments explores the relationship between humanity and urban landscapes. This study aims to forge a path towards greener, healthier urban futures, tailoring green designs to every corner of the city. This study examines architecture, landscapes, and urban planning in an effort to create a more green, more healthy urban environment. The Green Prescription for Urban Environments is to inspire and guide communities, legislators, and urban planners to prioritize humans and the environment.

Table of Contents

Abstract.....	06
1. Green Spaces.....	08
1.1. Literature Review.....	11
1.2. What Is a Green Space?.....	12
1.3. Green Impacts Human Health.....	14
1.4. Green Impacts the Environment.....	18
1.5. Green Impacts Human Health and the Environment.....	20
2. The Green Prescription.....	24
2.1. Historical Context.....	28
2.2. Ideology.....	30
2.3. Goals.....	34
2.4. Methodology.....	36
3. Case Studies.....	50
3.1. Parc Clichy-Batignolles - Martin Luther King.....	52
3.2. High Line.....	60
3.4. Fresh Kills Park.....	68
4. Design.....	76
4.1. Diagnosis for Monterrey.....	78
4.2. Diagnosis for Campo los Pinos.....	84
4.3. The Green Prescription for Campo los Pinos.....	88
5. Conclusion.....	96
Sources.....	98

Abstract

Urbanism is rising globally, causing impacts on both environmental and human health. The Green Prescription for Urban Environments is an approach that recognizes the connection between Earth's health and human health. By diagnosing and addressing urban health challenges, this prescription aims to restore environmental health while also improving physical, mental, and social well-being for individuals.

The Green Prescription acknowledges the health consequences created by urban living. Urban areas are being understood as ecosystems that link human and environmental health. The Green Prescription diagnoses and treats urban health issues by establishing a cycle in which people's well being supports to the restoration of environmental health, and vice versa.

Looking forward, the Green Prescription is a transformative approach, calling on communities, policymakers, and healthcare providers to prioritize green spaces in urban environments.

Green Spaces

1.1. *Landscape as Urbanism: A General Theory* by Charles Waldheim

Charles Waldheim's "Landscape as Urbanism: A General Theory" highlights the role of landscape architecture in shaping the modern city. Waldheim challenges traditional urban design approaches that prioritize built infrastructure over landscape elements. His thesis advocates for a modern way of viewing landscapes. According to his idea, landscapes are essential components of the urban environment, rather than being decorative spaces. According to Waldheim's view, landscapes have an active role contributing to ecological sustainability, social cohesion, and economic well-being within urban environments. He argues that landscapes can perform multiple functions simultaneously, such as supporting social and economic activities, providing ecological services, along with enhancing the overall urban aesthetic. He emphasizes the importance of perceiving landscapes as systems that change over time and interact with urban development.

One of Waldheim's significant contributions is his criticism of the split line between city and nature, and he proposes a hybrid approach where landscapes are integrated into the urban environment. This concept pushes urban designers to create interconnected urban forms that should accommodate both built environments and natural landscapes. As a result, encouraging stronger and sustainable cities.

Waldheim's emphasis on landscapes in the urban environment resonates strongly with me. Landscapes create environments that weave together the overall urban fabric. Landscapes add to the aesthetics of cities while aiding their functionality and livability through the mitigation of environmental impacts, and improving the overall quality of life.

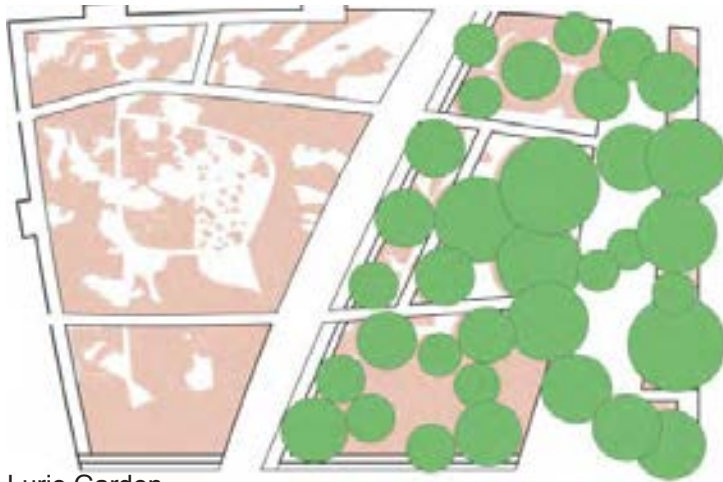
In conclusion, Charles Waldheim's "Landscape as Urbanism" provides the model for rethinking urban landscapes as functional and interconnected systems. Through his advocacy of including landscapes in urban planning, Waldheim challenges the urban planning norm and encourages the development of more sustainable and inclusive cities. His ideas continue to influence modern urban design practices, inspiring cities worldwide to recognize the importance of landscapes in urban growth.

1.2. What is a Green Space?

Firstly, the definition of a green space is a bigger area with lots of vegetation, such as trees, shrubs, flowers. A green space is seen to be a place of fun and recovery for people of all ages. Where visitors can have a retreat from the urban area surrounding them. Providing visitors with a place to walk, do sports, hang out with friends, even just to sit alone and enjoy the weather. When an urban green space offers much to do and see, especially in dense areas, it can be seen as a landmark of the city. This can potentially increase the amount of visitors in the area itself.

Green spaces provide space for people and nature alike. With enough space, a green area will allow for small animals and insects to live within it. Providing them with shelter, creating a small ecosystem within it, helping with biodiversity within the area.

The following are green spaces in the Chicagoland area which I've drawn and studied, highlighting the range of sizes and features in an urban green space:



Lurie Garden



West Ridge Nature Preserve



Montrose Point Bird Sanctuary



The Lurie Garden in Chicago known for its native plants, seasonal changes, and surrounding hedges
 PHOTO: JARED BARNES FROM MERISTEM



Montrose Point Bird Sanctuary, with a black-throated blue warbler visitor on the right
 PHOTO: HAVEN&LOFT REALTY (LEFT), OWEN DEUTSCH PHOTOGRAPHY (RIGHT)



West Ridge Nature Preserve provides retreat for animals and visitors alike
 PHOTO: ERIC ROJAS OF KALE REALTY

1.3. Green Impacts Human Health

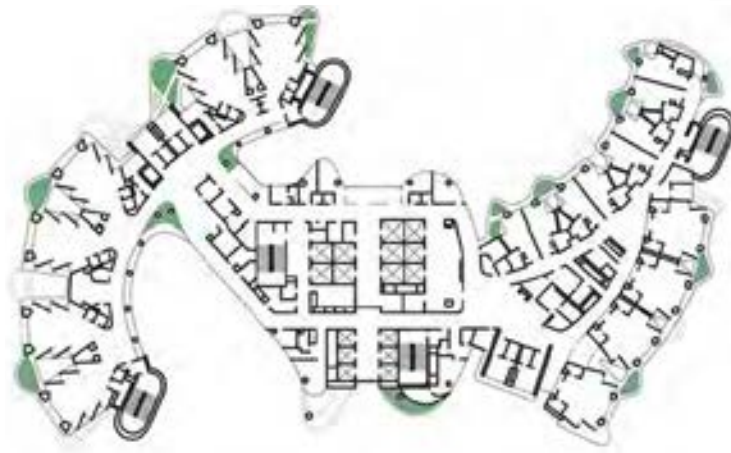
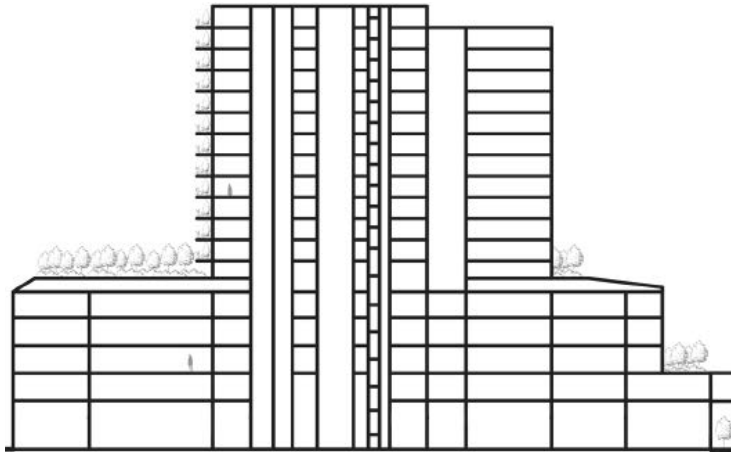
Improved Health Many studies have shown that green spaces are capable of increasing the physical and mental health of those who live nearby and utilize them. This has shown to be true, especially concerning the children and senior population. According to a study conducted in 2003, featured in the Journal of Epidemiology and Community Health, the percentage of green space inside a .6 mile radius and a 1.86 mile radius had a relation to overall general health. This study shows that the correlation between urban green spaces and general health appears to be high when concerning lower socioeconomic groups, the youth, and the elderly. A study conducted in 2004 at Deakin University displayed that green space is crucial for the mental and physical growth of children. The study conveyed that minors participated in physical activities depending on the proximity and access to green space.

Advanced Healing Over the years, many professional journals have featured articles about the effects of green spaces on individuals. A study featured in the 1984 Science Magazine featured a Pennsylvania hospital displaying that patients with a window view of a local city park experienced a higher recovery rate when compared to those with the view of the back of a building. Similarly, The Journal of Environmental Psychology featured a study in which views of urban green spaces (referred to as “restorative environments” by the author) are crucial for those living in urban environments. The article suggests that such environments have a critical role in restoring individuals’ well-being, rather than environments that don’t fully support their endeavors.

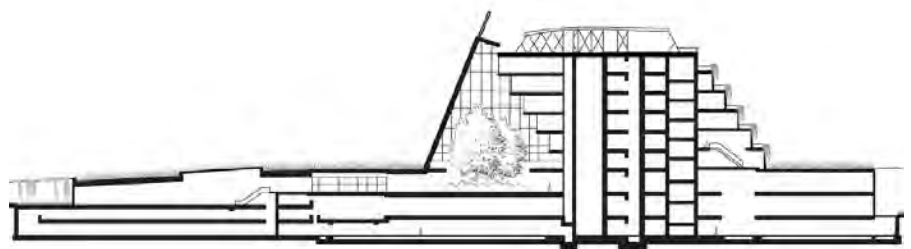
“Viewing nature has been repeatedly demonstrated to provide a range of benefits for human health and well-being. Benefits include reduced anxiety, reduced stress, shorter hospital stays, lower heart rate, and increased directed attention... It has repeatedly been shown that the sounds of nature such as wind, water, and animals, are preferred over anthropogenic sounds such as traffic, recreational noise, and industrial noise... Nature sounds have been used therapeutically to relieve stress.”

- A review in the International Journal of Environmental Research and Public Health

The following are plans and sections of healing centers that recognized and designed with the intent of green spaces to aid the healing process, drawn by me:



Ng Teng Fong General Hospital, CPG Consultants, Singapore 2015
PHOTO: CPG CONSULTANTS



Ospedale Dell'Angelo, Emilio Ambasz, Italy 2008
PHOTO: ALBERTO CIPRIANI ARCHITECT

1.3. Green Impacts Human Health

Sociability Strong community bonds among residents are crucial for the flourishing of neighborhoods. Studies have shown that parks can cause a rise in social connections in neighborhoods they're placed in. In 2012, the Management of Environmental Quality published an article explaining how parks can impact and increase social interactions within an urban area. This article explained that high density developments separates and isolates residents, resulting in the lack of sociability. The crowded built environment and human environment makes it difficult for people to interact. The Environment and Behavior Journal discussed a study indicating that seniors living inside a city with accessibility to green space communities have higher social ties within their neighborhoods. This emphasizes the value of green spaces across diverse social groups. Residents near green spaces have a greater sense of being involved in a community.

Activities and areas that increase sociability in a green space:

- Community Events
- Workshops and Classes
- Art Installations
- Interactive Games
- Group Exercising
- Walking Paths
- Cycling Paths
- Sports Areas
- Playground
- Picnic Areas
- Dog Park
- Community Garden
- Cafes or Food Vendors



Dolores Park, San Francisco
PHOTO: KATHERINE PAPERA/GETTY IMAGES



Primrose Hill, London, UK
PHOTO: SECRET CITY TRAILS



River Garden, Memphis, USA
PHOTO: ERIN MOSHER



Crown Fountain, Chicago, USA
PHOTO: MILLENNIUM PARK FOUNDATION



Kampa Park, Prague, Czech Republic
PHOTO: YOGA FEDERATION OF EUROPE



Parc de l'Espanya Industrial, Barcelona, Spain
PHOTO: COURTS OF THE WORLD

1.4. Green Impacts the Environment

Soil Conservation Green spaces tend to encourage soil conservation, regulating soil processes and overall health. Elevated soil organic matter (SOM) in parks improves soil quality through maintaining soil biodiversity and increasing its strength. Parks serve as reservoirs for soil organic carbon, retaining nutrients while reducing soil erosion and encouraging water filtration. Soil conservation does however get directly impacted by plant species diversity, litter, and management. Excessive maintenance can trigger soil health setbacks, declining stability and biodiversity.

Water Management Green spaces encourage water management through the filtration capabilities of soil. Soil acts as a natural filtration system, absorbing stormwater runoff and dissolving contaminants. Water infiltration in soil allows sediments to settle and diminish pollutants. Filtering pollutants benefits the environment since it prevents their infiltration into natural bodies of water. The

slowing of water in parks fosters more efficient filtration and sedimentation processes. Through buffers between impermeable and natural water bodies, parks help reduce the impacts of urbanization on water quality.

Biodiversity Preservation Urban green spaces play a crucial role in maintaining biodiversity in urban areas by providing a safe haven for various plant and animal species despite urban expansion. Urban parks offer vital habitats and pathways for several species of wildlife, supporting their migration across diverse urban areas. The amount and variety of wildlife supported by a park is influenced by various factors, including the park's size, the variety of habitats it contains, and the diversity of tree species. According to study findings, by creating suitable habitats and fostering connectivity between diverse natural areas, well-planned and maintained urban green infrastructure can improve biodiversity conservation.



Tanner Springs Park in Portland, USA
PHOTO: HERBERT DREISEITL



Kadma Biodiversity Park, Jarkhand, India
PHOTO: TEAM URBAN UPDATE

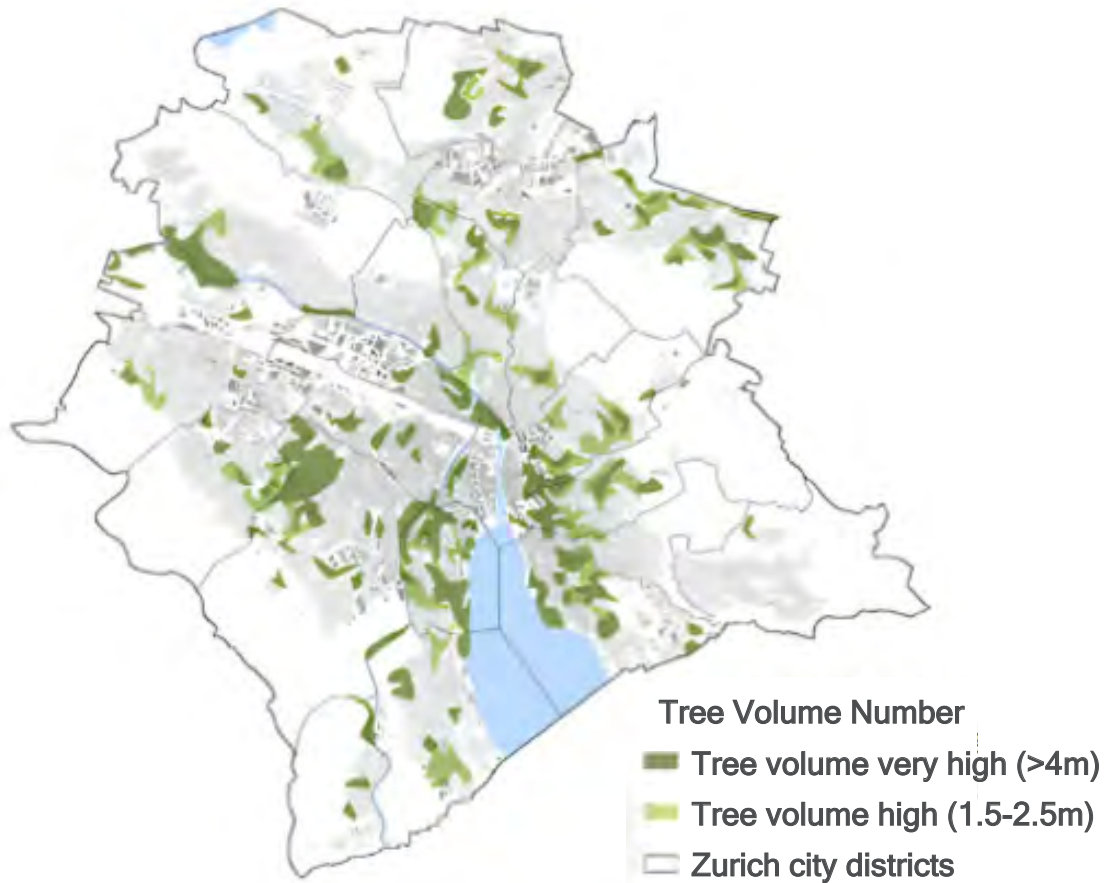
1.5. Green Impacts Human Health and the Environment

Carbon Sequestration In urban areas, parks play a vital role in carbon sequestration and in the reduction of the urban heat island effect. The urban heat island effect refers to the fact that urban regions have significantly greater temperatures than surrounding rural areas. This temperature difference is mostly caused by human activity and the built environment in cities. Urban regions typically include extensive expanses of concrete, asphalt, and buildings that absorb and retain heat while diminishing vegetative cover. Green spaces reduce carbon emissions from vehicles, buildings, and industries by absorbing CO₂ and other pollutants. The presence of various ecosystems inside parks improves their ability to hold onto carbon, as different plant species have varying carbon absorption and storage capabilities. Cities can improve their resilience to climate change by fostering biodiversity within urban parks, as well as providing wildlife habitat. Furthermore, parks provide shade and lower surface temperatures, resulting in healthier and more habitable urban surroundings.

Heat Relief Humans have left their mark on the Earth and its environment, and it's now important for cities to implement green practices. Green spaces play a crucial role to ensure the health of the environment

through “invisible” services. In 2011, an issue in the *Journal of Environmental Protection* demonstrated the benefits of urban green space. Parks of diverse sizes stand out, along with specialized and institutional green areas such as cemeteries, schools, and swimming pools, as well as the surrounding residential and work settings. A park starting at roughly 2.5 acres (1 hectare) in size, the green spaces within the park facilitates the exchange of cooler air with the adjacent neighborhoods. This serves as an oasis for retreat and relaxation.

Green spaces play a role in the heat relief systems in the urban environment. A study conducted in Zurich demonstrated that during the day, additional trees on lawns reduce the felt temperature by 40°F as the median and 48°F as the maximum value. The difference is particularly significant in this scenario because the evapotranspiration process from the lawn causes a larger decrease in temperature compared to having a tree over asphalt. In this case, the temperature drop averages 36.14°F, with a maximum reduction of 40.1°F. These green spaces nestled within the urban heat island act as cool spots offering residents, local workers, and animals with sanctuaries for relief during periods of intense heat.



Felt Temperature Night (air temp.)	Median (°F)	Max (°F)	Effective Range (m)
Tree on lawn	24.44 30.92	16.34 29.84	7-20 2-5
Grass (rather than asphalt)	25.52 31.28	20.12 30.74	2-4 2-3
Pergola on asphalt	25.38 31.46	21.20 30.56	2-3 2-3
Water area	21.92 31.46	18.32 29.84	3-6 2-4
Tree on asphalt	27.86 31.64	23.90 31.10	7-20 2-3
Retention room instead of lawn	30.20 31.82	29.48 31.64	5-18 5-10

Map displaying high tree volumes in Zurich (above), graph of temperatures in locations in Zurich (below)
FIGURES FROM: BEFTER21 CITY PARKS

1.5. Green Impacts Human Health and the Environment

Noise Pollution Urban areas tend to be impacted heavily by noise pollution, stemming from transportation, industrial areas, and other human activities. Trees help combat this issue through the absorption and obstruction of sound vibrations. Intentionally placed greenery has the potential to reduce traffic noise by 5-10 decibels, reducing the perceived loudness. Evergreen and broadleaf trees with dense foliage are efficient at sound absorption, though clustered or staggered tree arrangements of any species have also been proved to ensure year-round insulation against these noises. Mature trees offer immediate results when strategically planted near noisy areas.

Air Quality Connected forested areas, fields, and meadows are crucial zones for generating cold air, providing oxygen-rich, dust-free, and minimally polluted air. The interconnection of green spaces is achieved through the planting of trees along pathways and roads, along with periodic resting spots, in the form of small parks, open areas, and even gardens. To increase effectiveness, it's recommended to incorporate a diverse range of open spaces and vegetation structures, consisting of large, shady trees, expanses of open fields, and water features, to provide the most climate resilience. Direct radiation and nighttime heat emissions are combated by shaded outdoor areas and building facades. Additionally, the

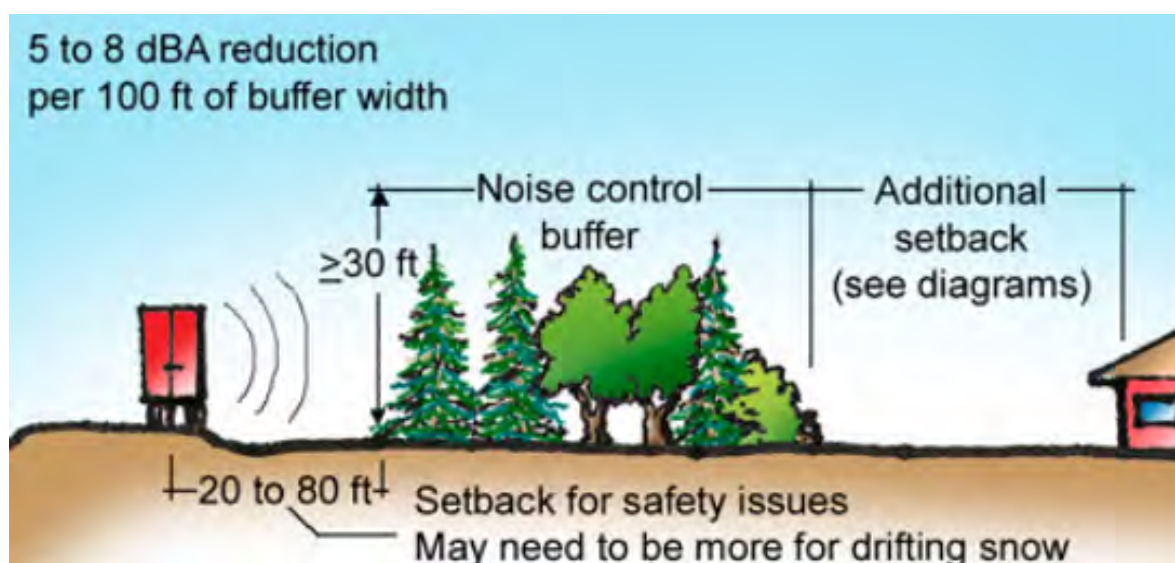


Diagram representing trees as a buffer from industrial noise pollution
DIAGRAM FROM: NATIONAL AGROFORESTRY CENTER

coolness resulting from vaporization from vegetation also reduces the bioclimatic stress.

Trees play a vital role in improving air quality through capturing particulate matter (PM) from the atmosphere. While PM pollutants are known to be harmful to human health, they can also be harmful to vegetation and cause growth reduction. Particle deposition to tree canopy is dependent on factors such as particle capture efficiency (C_p) and deposition velocity (V_g). C_p varies based on wind speed, particle size, and tree properties. Leaf anatomy, including leaf area and structural wax, affects particle deposition. C_p values vary across different tree species, ranging from 0.03% to 0.38%. Various methodologies are used for calculating C_p , which is further influenced by the anatomical and physical traits of leaves.

In a study involving Coniferous Scots pine and three broadleaf species, Coniferous

Scots pine had the highest particle capture efficiency (C_{ptot}). Among these tested species, Coniferous Scots pine has a significantly greater C_{ptot} than broadleaved species. Previous studies also found coniferous species like Corsican pine to have high capture efficiency. Pubescent birch was the most efficient particle collector among broadleaved species.



Scots pine / *Pinus sylvestris*
PHOTO: MEYE



Pubescent birch / *Betula pubescens*
PHOTO: MEYE



Scots pine leaves (above), Pubescent birch leaves (below)
PHOTO: THE GROVE (above), ORTIS (below)

Green Prescription



Central Park, New York, USA
PHOTO: ZEESHAAN SHABBIR



2.1. Historical Context

Many North American cities have been transformed from the post-Fordist models of production to leisurely environments. These cities have become themed environments for tourism, recreation, and destination. In the traditional city, most North Americans spend their time in built environments known for their density, easy access for vehicles and other transportation, and public spaces that have been heavily characterized by vegetation. However, residents and tourists alike seek spaces away from the density and the chaos of the modern city, which tends to be found in green spaces.

A great example of this is Central Park, in New York City. In 1853, the New York State Legislature set aside land, consisting of rocks, swamps, and marshlands, to be turned into a public space for New Yorkers. The park's 834 acres were utilized by designers Frederick Law Olmsted and Calvert Vaux to create a range of landscapes and experiences, consisting of lawns, woodlands, streams, and lakes, that are accessible through different paths inside the park. The intent of the park was to offer residents a countryside experience, where visitors can escape the stresses of urban life and to be one with nature and with fellow New Yorkers. A main goal of the design was to provide a space where all people can come together, regardless of their backgrounds. As of 1963, Central Park's importance was recognized and



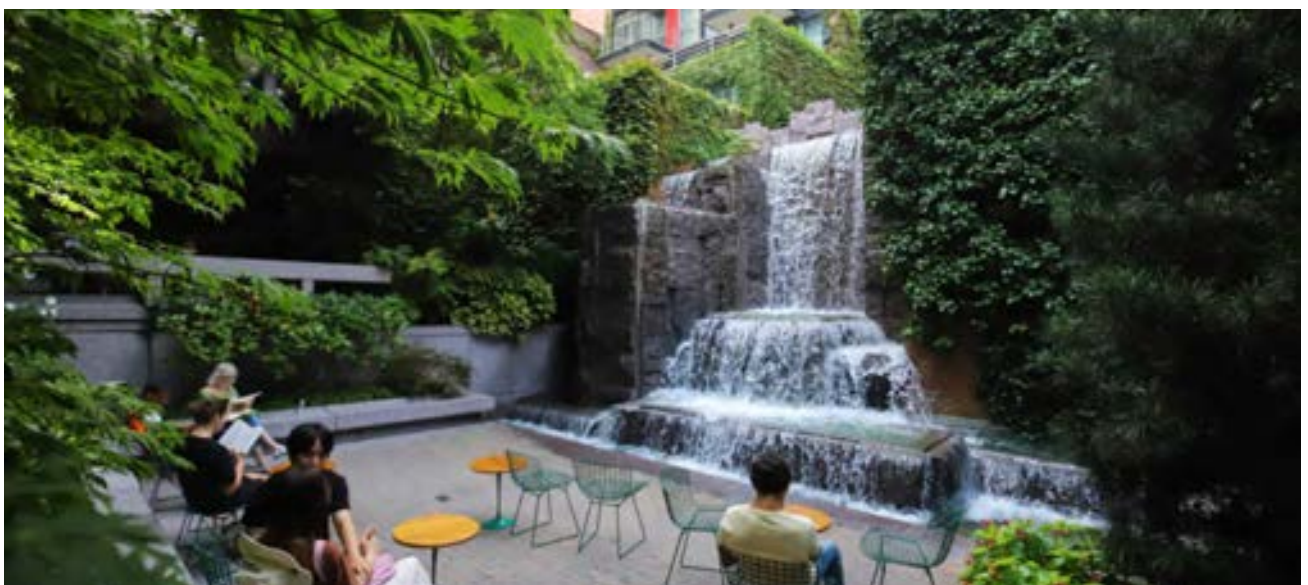
Central Park, New York, USA
PHOTO: TEAM URBAN UPDATE

the park was deemed a National Historic Landmark. Throughout the years, the park has been enhanced with different spaces, such as skating rinks, a zoo, formal gardens, monuments, along with concert/theater venues. Central Park continues to influence the development of urban parks nationally and worldwide.

At the start of the twenty-first century, landscape has transformed into a modern practice. The reappearance of landscape can be attributed to the increase of environmentalism and heightened global ecological awareness. Along with the increase in tourism and the needs of a region to sustain agency. Landscape has transformed from simply drawing trees into an interest in vegetation and detailed site-planning.

In the contemporary urban environment, landscape is seen as a constructed medium, similar to architecture itself. Landscape, urbanism, and architecture have transitioned into a shared form of practice, leading into the term “landscape urbanism.” This new discipline was displayed in the Landscape Urbanism symposium and exhibition in 1997, conceived by Charles Waldheim. It proposes a merging of architectural, urbanism, and landscape practices and disciplines into one. The unity embraces the differences within the ideological, programmatic, and cultural aspects that are associated with the terms “landscape” and “urbanism.”

Landscape urbanism differs from earlier attempts to speak about urban sites as landscapes, differentiates it from attempts to situate landscapes in the city. Traditionally, cities and landscapes are discussed from different lenses. From this perspective, cities are seen as having densely populated areas, efficient transit, extensive infrastructure, and development that affects traffic, pollution, and stress levels among locals. On the other hand, landscape features including gardens, esplanades, parks, greenways, and street trees are frequently perceived as havens from the effects of urbanization.



Greenacre Park, a pocket park in New York City, USA
PHOTO: SAI MOKHTARI

2.2. Ideology

Historically, urban planning has often taken more attention in urban development, rather than natural features. As a result, cities have been referred to as “concrete jungles” because of their concrete landscapes with little to no green space, contributing to poor air quality, heat island effects, along with poorer well-being amongst residents.

The Green Prescription is an approach to urban planning that integrates green spaces into the built environment. Similarly to Charles Waldheim, The Green Prescription advocates for landscapes to be a standard in the design of cities. At its core, this ideology seeks to transform unused and vacant infrastructure and land into functional green spaces. This concept enhances the environment while boosting public health at the same time.

Similarly to Charles Waldheim’s advocacy, The Green Prescription emphasizes that landscapes should be a standard in city design, recognizing that green spaces offer vital functions beyond merely “greening” the city. These green spaces provide benefits to both the surrounding people and environment. This approach could be viewed as opportunistic, using the potential of underutilized landscapes to provide specific functions, such as gardens, pedestrian pathways, even community farms. In order to implement this plan successfully, green spaces should be integrated into the urban fabric. As a result, improving functions such as mobility, stormwater management, biodiversity, and addressing social issues related to recreation and leisure.



The proposal of a parking lot to a green space in London
 PHOTO: EAST LONDON WATER WORKS PARK

2.2. Ideology

Philadelphia is a prime example of how transforming vacant lots into green spaces can have positive benefits on both the people and the environment. JAMA Network Open published a study that highlights the relationship between the restoration of lots in Philadelphia and the mental health of city residents. Greening vacant lots in Philadelphia has been associated with a decrease in depressive symptoms and an improvement in general mental health among neighborhood inhabitants. “Vacant lot greening is a very simple structural intervention that’s relatively low-cost and that can have a potentially wide or broad population impact,” says Dr. Eugenia South, co-author of this study and an assistant professor at the University of Pennsylvania’s Perelman School of Medicine. She also explains that “performing simple interventions to the neighborhood environment has an impact on health.”

The Pennsylvania Horticultural Society (PHS) has been leading this change through its Philadelphia LandCare program. This globally acclaimed program improves environmental health and the quality of life through the transformation of neglected lots into community assets. PHS’s efforts are especially important in low and moderate income communities, in which vacant lots are more likely due to past disinvestment. Through hiring minority-owned businesses within the community to clean and maintain these spaces, PHS not only restores the neighborhoods but also creates employment opportunities and aids the local economy. Research has shown a 40% decrease in feelings of depression, a 51% decrease in feelings of worthlessness, and a 29% reduction in gun violence in neighborhoods with LandCare lots. “This is not an end-all and be-all treatment for depression in any way,” claims Dr. South. Though it is a step towards better mental health.



Before the restoration of a vacant lot in Mantua, Philadelphia
PHOTO: HOLLY GALLAGHER

A notable project that NWF led was the installation of pollinator gardens at 50 carefully chosen sites in Philadelphia. The group determined the best sites for planting 13 native plant species through the use of GIS and field research. As part of the effort, varying in design and size, gardens were installed and maintained as part of the initiative. The project included field research to evaluate the environmental effects of these gardens.

Research led by expert Dr. Chris Swan of the University of Maryland Baltimore County, revealed that 8 months later the pollinator gardens were much more environmentally resilient than the lots surrounding them. The native gardens showed an increase in the ability to retain soil moisture and a decrease in soil compaction, emphasizing their impacts to retain stormwater. This improvement in soil condition, compared to nearby turf grass, the nearly instantaneous environmental benefits of such greening initiatives.

The success of these efforts further demonstrate the importance of green spaces. According to the study, residents who live near green lots benefited significantly more mentally in comparison to those living near lots that were only cleaned. According to South, “the green space in and of itself is important,” pointing out the different ways in which green spaces improve well-being, including increased social connections along with stress relief. The conversion of an abandoned place into a cared for location is essential for promoting a sense of belonging.

Repurposing unused lands into green spaces is an urban design strategy that addresses environmental issues while also promoting social cohesion, safety, and mental health. With urbanization and climate change putting more and more strain on cities, Philadelphia’s example demonstrates how strategically greening vacant land can be powerful for enhancing urban life.



After the restoration of a vacant lot in Mantua, Philadelphia
PHOTO: NATALIE COHEN

2.3. Goals

1. Enhancing Public Health

The accessibility of green spaces is aimed to improve health. As discussed in the previous chapter, green spaces and an overall proximity to nature helps lower stress, encourage physical activity, and improve well-being. The Green Prescription seeks to provide outdoor recreation and relaxation by transforming abandoned infrastructure into accessible parks, gardens, with the potential for other green spaces.

2. Increased Community Interactions

When designed well, green spaces serve as a congregation space. Green spaces promote social interactions through providing places for people to gather, interact and engage in local activities. The Green Prescription aims to enhance social lives and foster a sense of community through the development of easily accessible and welcoming spaces.

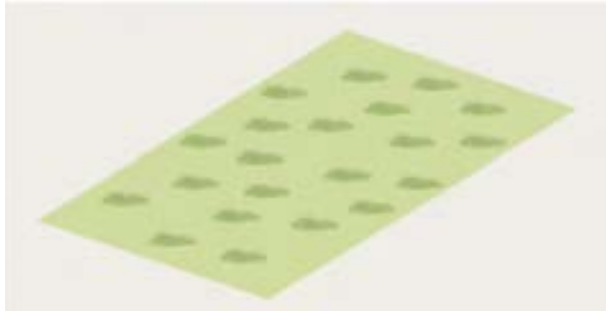
3. Mitigating Environmental Impacts

Urban areas frequently experience environmental concerns such as the heat island effect, where urban areas have become much warmer than their surrounding areas. Green spaces contribute to the relief of these problems by managing stormwater and enhancing air quality. The Green Prescription aims to tackle these problems through the use of strategically placed native plants.

4. Promoting Biodiversity

By incorporating green spaces in urban environments, habitats are created for a variety of species, helping increase biodiversity within city boundaries. The Green Prescription protects and improves ecosystems by transforming vacant land into a natural green space, bringing nature back into the city.

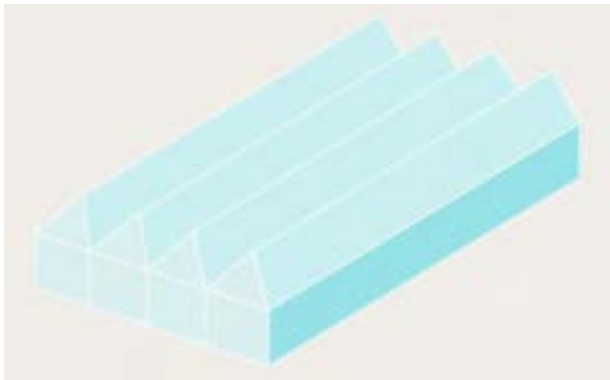
The Green Prescription aims to create green spaces that are both desired and needed by communities, including spaces such as:



Lawn



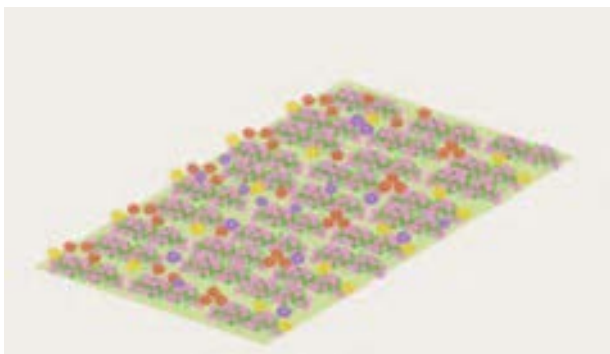
Forest Pathway



Greenhouse



Urban Farm / Community Garden



Native Plant Garden



Plaza

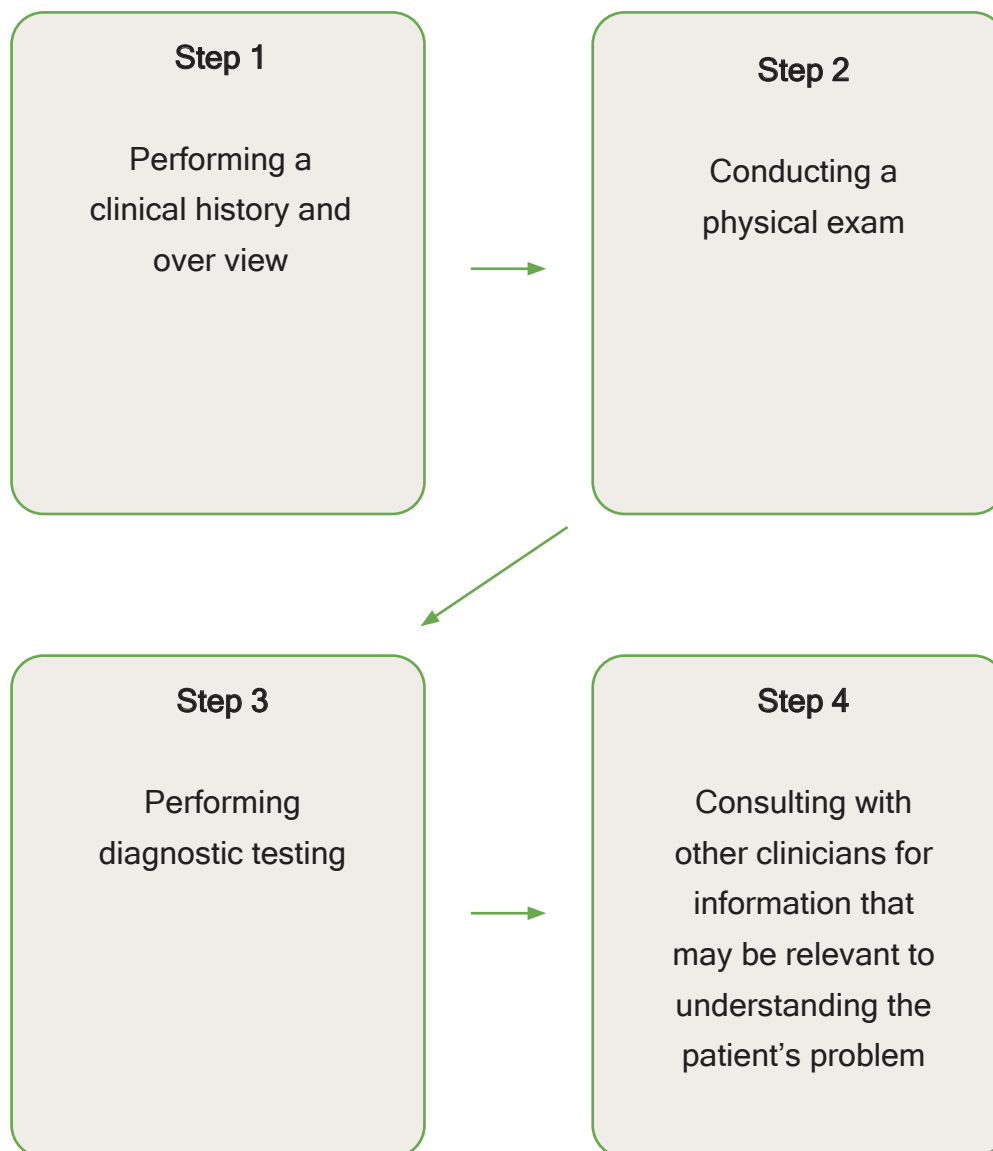
2.4. Methodology: Diagnosis

Differential Diagnosis

In healthcare, a differential diagnosis is a method of analysis that distinguishes a particular disease or condition from others that present with similar clinical features

The goal is to arrive at the correct diagnosis for the patient by ruling out other possible explanations for the symptoms.

Differential Diagnosis Steps



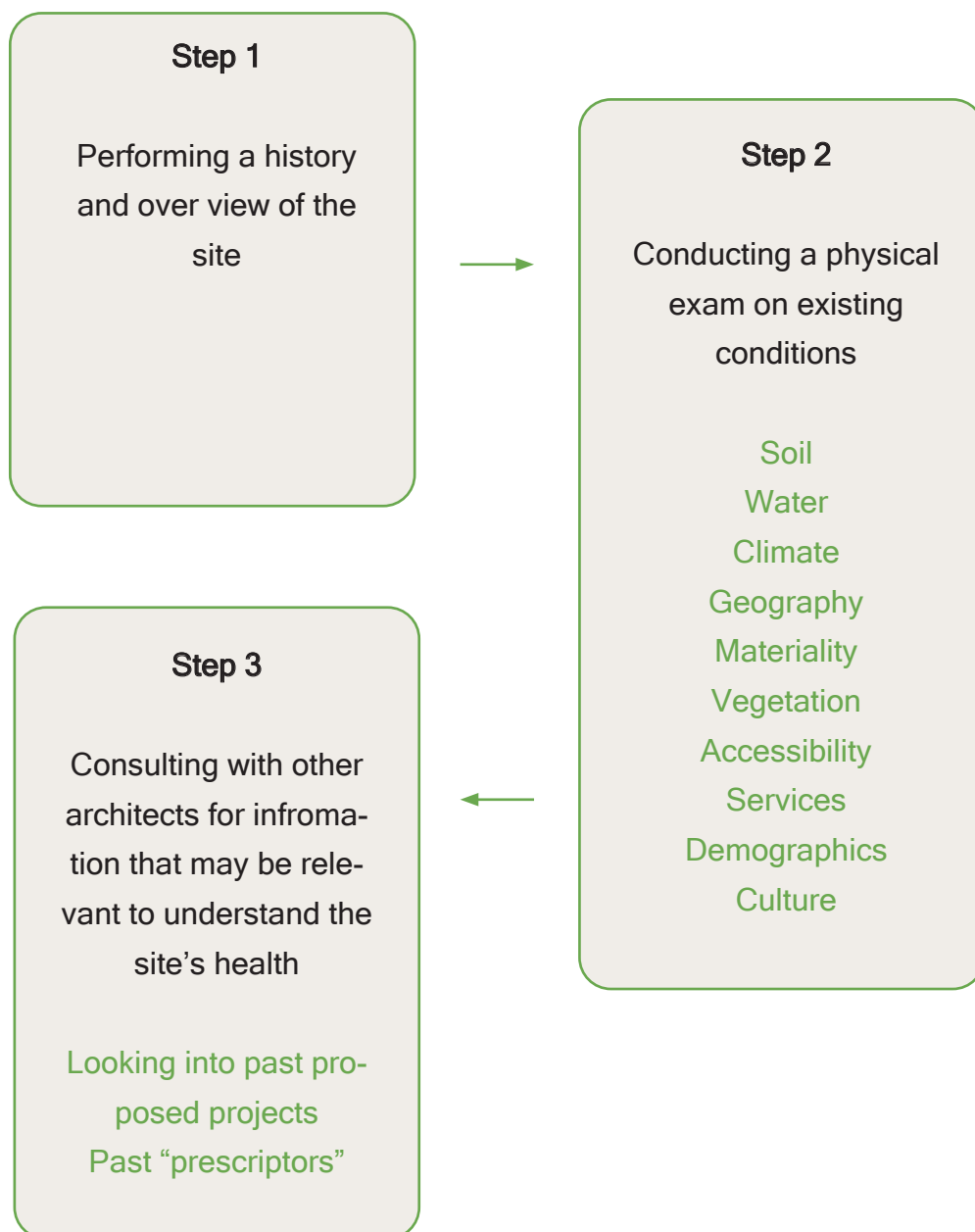
It is important to note that the diagnosis process needs sufficient and accurate information, if it isn't, the process can be repeated

Site Differential Diagnosis

In the Green Prescription, a diagnosis is a method of site analysis that distinguishes what can be done to a site to make it more human friendly, along with aiding the environment

The goal is to arrive at the appropriate analysis of the site to ensure the correct site design, suiting the needs for the visitors and the environment

Site Differential Diagnosis Steps



Utilizes research to assess urban environments and diagnose coniditons affecting human health and the quality of life

2.4. Methodology: Prescription Manual

Prescription Manual

The Green Prescription prescribes various design elements for the affected site based on the specific needs of the site. The goal is to contribute to the restoration and enhancement of human and environmental well-being in urban settings.

<u>Design Elements</u>	<u>Methods</u>
Shade and Shelter	• Shading structures • Trees • Seating areas
Connectivity	• Well connected paths • ADA features • Proximity to local residents
Vegetation	• Native plants • Edible plants • Fruit bearing plants • Trees
Water Management	• Rainwater gardens • Bioswales • Retention ponds • Sloped surfaces
Sustainable Features	• Permeable paving • Renewable materials • Energy efficient lighting

General Impacts

- Relief from the sun
 - Encourage longer stays
 - Promote social interaction
 - Reduce stress
-

- Encourages physical activity
 - Social interactions
 - Reducing the reliance on cars
-

- Can support biodiversity
 - Provide habitats
 - Air purification
 - Temperature regulation
-

- Can improve water quality
 - Mitigate flooding
 - Potential water habitats
-

- Reduce environmental impact
- Long-term span

2.4. Methodology: Prescription Manual

Shade and Shelter

The Green Prescription can enhance a site through prescribing shade and shelter.

Various methods of this design element includes:

Shading Structures

Shading structures are crucial to protect visitors from the weather, especially from the sun and rain. These structures provide heat relief, encourage longer stays, and can even enhance overall aesthetics. Types of shading structures are:

- Gazebos
- Pavilions
- Canopies
- Added walls

Trees

Trees provide natural shade and shelter. They help block the sun, bring heat relief, promote biodiversity, improve air quality, and reduce stress.

Seating Areas

Seating spaces, especially with shade, invite and welcome visitors to relax, and socialize. Seating spaces provide visitors encourage longer stays and promote social interactions. They can be incorporated in the design through:

- Benches
- Picnic tables
- Curved seating
- Integrating the seating into landscape



Japanese-styled gazebo (left), Superior shade at University of West Georgia
 PHOTO: STAN FAIRBROTHER (LEFT), SUPERIOR RECREATIONAL PRODUCTS (RIGHT)



Prospect Park in Brooklyn, New York City, USA
 PHOTO: CLARISA DIAZ



Hyperlane Linear Sky Park in Chengdu, China (left), Atlanta Financial Center in Atlanta, GA, USA (right)
 PHOTO: BING LU (LEFT), WAUSAU TILE (RIGHT)

2.4. Methodology: Prescription Manual

Connectivity

The Green Prescription can enhance a site through connectivity.

Various methods of this design element includes:

Well-Connected Pathways

Well-connected pathways promote a smoother flow of movement throughout the site, providing overall easier use. Connected paths attract more and frequent visitors, increasing physical activity, and boost social interactions between visitors. Well-connected pathways can be demonstrated through:

- Hierarchy of pathways
- Materiality
- Connection to outer street

ADA Features

ADA accessibility makes the site inclusive for everyone. ADA design elements throughout the site ensure access and use amongst all visitors. This can contribute to more overall visitors, increased visitor happiness, along with boosted mental and physical health amongst visitors.

Ways to make a site more accessible are:

- Accessible pathways
- Accessible entry and exit
- Accessible amenities and seating
- Tactile paving
- Audible signals

Proximity to Residents

A site's location in relation to residential areas impacts its accessibility and usage. Sites that are reachable can encourage regular use, reduce the reliance on cars, foster a sense of community and social interactions, and encourage active lifestyles. Proximity can vary through the site location as well as its access to the nearest street.



Southern Parklands Framework in Sydney, Australia
PHOTO: TYRRELL STUDIO



Tongva Park + Ken Genser Square in Santa Monica, California, USA (left), tactile pavement (right)
PHOTO: TIM STREET-PORTER (LEFT), SHUTTERSTOCK (RIGHT)



Pocket park adjacent to residential apartments in Athens, Greece
PHOTO: GTP HEADLINES

2.4. Methodology: Prescription Manual

Vegetation

The Green Prescription can enhance a site through vegetation.

Various methods of this design element includes:

Shrubs and Trees

As previously mentioned, trees and shrubs can be incorporated into a design to provide shade, ultimately helping cool down the site, absorb carbon and noise, support well-being, and add to the visitor experience.

Native Plants

Native plants have changed and adapted to specific regions over time with regards to the soil type, climate, and animals within that area. Compared to non-native plant species, this makes them require less resources such as water, fertilizer, and pesticides. Native plant gardens can help teach the community about the importance of supporting them, increasing biodiversity, enhancing temperature regulation, and reducing overall maintenance costs. Native plants can consist of:

- Wildflowers
- Grasses
- Shrubs
- Perennials



Bellevue Botanical Garden native plants in Bellevue, Washington, USA
PHOTO: KING CONSERVATION DISTRICT

Edible and Fruit Bearing Plants

Edible plants, like herbs, leafy greens, and vegetables, enhance and create a unique and enticing landscape for both humans and animals alike. Edible plants promote healthy eating, community engagement, and provide resources for the community (when properly cared for). These type of plants not only support humans, but they support wildlife through providing fresh and local food for them. Edible and fruit bearing plants can be integrated into the design by:

- Community garden
- Edible landscaping
- Incorporating fruit trees



Edible plant garden (left), Citrus Garden in the Botanical Garden of Chania, Crete, Greece (right)
PHOTO: LIVING COLOR GARDEN CENTER (LEFT), MARTIN RUTLAND (RIGHT)



Zaryadye Park in Moscow, Russia, displaying the use of trees and shrubs
PHOTO: ADOBE STOCK

2.4. Methodology: Prescription Manual

Water Management

The Green Prescription can enhance a site through water management.

Various methods of this design element includes:

Stormwater Gardens

Stormwater gardens, also referred to as rainwater gardens, are depressed gardens filled with vegetation, that are intended to absorb rainwater through the plants and soil. They are designed with plants that can tolerate the precipitation of the area such as native plants, which were previously mentioned. This design method can reduce flooding, filter out water pollutants, provide habitats, and add to the aesthetic of a landscape while also being functional.

Bioswales

Bioswales are designed to slow rainwater by directing it through a path leading to a stormwater garden, or any other area that can absorb the excess water. They have the same impacts as stormwater gardens, however with the intention of controlling the water flow.

Retention Ponds

Retention ponds are depressed and open areas with the intention of water seeping through the ground. Retention ponds capture larger bodies of water, reducing the risks of flooding, enhancing water quality, providing habitat, and they are often seen as an aesthetic feature.

Sloped surfaces

Similar to bioswales, sloped surfaces are intended to direct excess water to specific collection points, helping prevent flooding and providing effective water drainage.



Stormwater garden in Xuhui Runway Park in Shanghai, China
PHOTO: SASAKI



Bioswale in the Southeast Community Center in San Francisco, California
PHOTO: MERISTEM DESIGN



Retention pond in Farfield County, Connecticut, USA (left), sloped garden in Buckie, Scotland (right)
PHOTO: LANCASTER EAGLE-GAZETTE (LEFT), CUTHBERTS BRAE GARDEN (RIGHT)

2.4. Methodology: Prescription Manual

Sustainable Features

The Green Prescription can enhance a site through sustainable features.

Various methods of this design element includes:

Permeable Paving

Permeable paving are materials designed to allow water to pass through it, aiding the absorption of water into the ground below. Permeable paving can be created with materials such as:

- Porous asphalt
- Pervious concrete
- Interlocking pavers
- Gravel

Renewable Materials

Renewable materials are taken from resources that can naturally replace themselves. In green spaces, these can be used for benches, playgrounds, patios, art, and shading structures. Renewable materials can help reduce carbon footprints, increase durability, and support sustainable ideologies, while adding to the aesthetic. Common renewable materials are:
Bamboo

- Reclaimed wood
- Cork
- Recycled plastic
- Recycled steel

Energy Efficient Lighting

Energy efficient lighting can come from LED, wind-powered, and solar-powered lights. A great example of these are the lights installed along the IIT parking lot. These types of lighting lower energy costs, help minimize light pollution, and reduce the emission of fossil fuels.



From left to right: porous asphalt, pervious concrete, permeable interlocking concrete pavement, grid pavement
PHOTOS: SUSTAINABLE MANAGEMENT OF TRANSPORTATION INFRASTRUCTURE PROJECTS



From left to right: bamboo, reclaimed wood, cork
PHOTOS: ADOBE STOCK



Solar and wind powered lights in China
PHOTO: WADE SHEPARD

Case Studies



Parc Clichy-Batignolles – Martin Luther King



PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE



PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21

3.1. Parc Clichy-Batignolles - Martin Luther King

Location: Paris, France

Area: about 25 acres

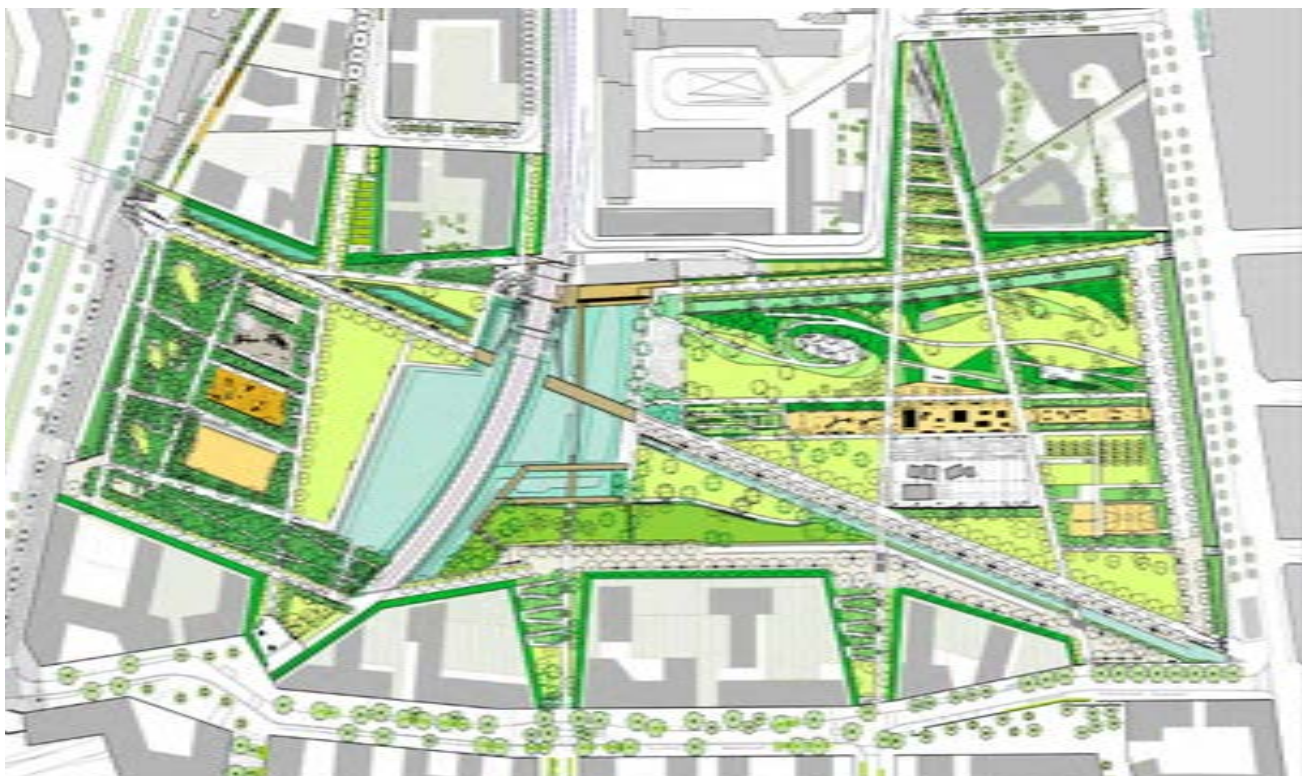
Completed: 2021

Architect: public space

Parc Clichy-Batignolles - Martin Luther King is made up of about 25 acres and offers areas for exploration and relaxation. Whether visitors are into sports, drawn to bodies of water, interested in urban agriculture, or its gardens, the overall park is designed to be suited for people of different interests. Additionally, its paths create a peaceful atmosphere for walks, jogs, picnics, or simply spending time within nature. Named in honor of Martin Luther King Jr., the park is a tribute to his legacy of equality and justice.

The park features a sustainable watering system by collecting rainwater in storage tanks to water plants. Additionally, a wind turbine powers the water distribution system throughout the park.

The project design began in 2003, with construction itself began in 2005. After years of planning and development, the park itself was completed in 2021.



Parc Clichy-Batignolles - Martin Luther King master plan
PHOTO: LES VICTOIRES DU PAYSAGE



Parc Clichy-Batignolles - Martin Luther King before
PHOTO: ATELIER JACQUELINE OSTY



Parc Clichy-Batignolles - Martin Luther King envisioned after
PHOTO: ATELIER JACQUELINE OSTY

3.1. Parc Clichy-Batignolles - Martin Luther King

Location: Paris, France

Area: about 25 acres

Completed: 2021

Architect: public space

- 
- 2003**
City of Paris called for a transformation of Clichy-Batignolles with the potential of hosting the Olympic Village for Paris's bid for the 2012 Olympics
 - Team selected: François Grether, Jacqueline Osty, and OGI engineer firm
 - 2007**
1st phase of the park opened to the public
 - 2008**
Bertrand Delanoë, mayor of Paris from 2001 to 2014, chose the name to mark the 40th anniversary of Martin Luther King Jr.'s assassination
 - 2014**
2nd phase of the park opened to the public
 - 2020**
Park was fully completed and opened to the public



Parc Clichy-Batignolles - Martin Luther King along the tracks, displaying its water features
 PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21



Strolling through a path in Parc Clichy-Batignolles - Martin Luther King
 PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21



Parc Clichy-Batignolles - Martin Luther King pathway and scenery
PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21



Parc Clichy-Batignolles - Martin Luther King basketball court
PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21



Parc Clichy-Batignolles - Martin Luther King skate area
PHOTO: MARTIN ARGYROGLO PHOTOGRAPHE, 2020-21



High Line



PHOTO: CONDE NAST TRAVELER



PHOTO: IWAN BAAN, 2011

3.2. High Line

Location: New York City, New York, USA

Area: 1.45 miles

Completed: 2009 but is continuing to get worked on

Architects: James Corner Field Operations, Diller Scofidio + Renfro

The High Line, overseen by the Department of Parks & Recreation once served as the West Side's industrial railway. Constructed in the 1930s for freight trains, this elevated steel structure spans 1.45 miles. Its final train journey occurred in 1980. Spanning from Gansevoort Street in the Meatpacking District through the West Chelsea gallery neighborhood to 34th Street near the Jacob Javits Convention Center, it now serves as a public park following an open competition held in 2003 to repurpose the existing infrastructure.

The High Line officially opened to the public in 2009 and continues to be developed.



Overview plan of High Line
PHOTO: ARQUITECTURA VIVA



The High Line when it was an elevated train track
PHOTO: LEARNING WITH EXPERTS



The High Line before it was transformed into a park
PHOTO: HIGH LINE



The High Line now
PHOTO: IWAN BAAN

3.2. High Line

Location: New Yory City, New York, USA

Area: 1.45 miles

Completed: 2009 but is continuing to get worked on

Architects: James Corner Field Operations, Diller Scofidio + Renfro

1924

New York City's Transit Commission ordered for the removal of street-level tracks and established an elevated rail line

1933

- The first train ran on the "West Side Elevated Line," now called the High Line
- Within the year, the line was fully operational and used to move meat, dairy, and produce

1960s-1980s

Train use declined, called for demolition of the structure

1983

Ideas for the structures reuse began

1991

Five blocks of the structure were demolished

1999

Inspired by the plants that were growing on the abandoned track, Friends of the High Line established to push for the preservation and reuse of the space

2003

Friends of the High Line hosted a competition for proposal ideas, receiving 720 ideas

2004

James Corner Field Operation, Diller Scofidio + Renfro, and Piet Oudolfwere chosen to be the team that designed The High Line

2009

The first section of The High Line was finished being built and opened to the public, and The High Line continued to get worked on till 2023



Aerial view of the High Line
PHOTO: IWAN BAAN, 2011



Birds-eye detail view of elevated steps in the High Line
PHOTO: CAMERON DAVIDSON



A walking path in the High Line
PHOTO: IWAN BAAN, 2011



Different environments on the High Line
PHOTO: BRITTANY PRETONELLA



Visitors enjoying the view from the High Line
PHOTO: TAGGER YANCEY IV



Freshkills Park





PHOTO: FIELD OPERATIONS

3.3. Freshkills Park

Location: Staten Island, New York, USA

Area: 2,200 acres

Completed: scheduled to be fully completed in 2036

Architect: James Corner Field Operations

Over 1,000 acres of former landfill and 450 acres of wetland makes up Freshkills Park. It has the space to allow for a variety of public spaces and amenities. Research, art, education, and recreational activities such as birdwatching, hiking, biking, running, even kayak tours are among the main programs to be offered.

Major design goals for the park have been making the site accessible to the public and emphasizing its features, including creeks, marshes, meadows, and views of downtown Manhattan.

Freshkills Park is home to several ecosystems with a wide range of wildlife. Many plants and animals have flourished since the landfill was closed, and the park's design is intended to further their growth. The capped landfill mounds were then covered with native grasses, creating a home for the rare grassland birds that now nest there. As a result, Freshkills Park has grown into one of the region's largest grassland environments.

Freshkills Park aims to be completed and open to the public in 2036.



Freshkills master plan
PHOTO: ARCHITECTURAL RECORD



Isle of Meadows, Fresh Kills, 1933
 PHOTO: STATEN ISLAND MUSEM



Fresh Kills Landfill, 1961
 PHOTO: DSNY PHOTO ARCHIVE



Draft Master Plan
 PHOTO: NY DEPARTMENT OF PARKS AND RECREATION

3.3. Freshkills Park

Location: Staten Island, New York, USA

Area: 2,200 acres

Completed: scheduled to be fully completed in 2036

Architect: James Corner Field Operations

- 
- 1948**
Robert Moses and the City of New York created Fresh Kills Landfill
 - 1990**
Landfill closure ordered between New York State Department of Environmental Conservation and the City of New York
 - 1996**
 - State law passed, requiring the landfill to reject solid waste by December 31, 2001
 - South Mound was sealed
 - 1997**
North Mound was sealed
 - 1999**
The Department of City Planning established an International Design Competition Organizing Committee to sponsor a master plan for Freshkills Park
 - 2001**
 - The landfill accepted its final solid waste
 - The City of New York launched the International Design Competition for Freshkills Park
 - Three finalist teams were selected by a jury to compete for the role
 - 2003**
The Department of City Planning established an International Design Competition Organizing Committee to sponsor a master plan for Freshkills Park
 - 2006**
The draft master plan was released
 - 2008-**
Construction for the park began and will continue for about 30 years, the North Park area opened in 2023 to the public



Visitors cycling and walking through Freshkills Park
PHOTO: NYC PARKS



Visitors kiting and relaxing in the park
PHOTO: NYC PARKS



Freshkills North Park, completed in 2023
PHOTO: AIA NEW YORK



Freshkills kayak tour
PHOTO: FRESHKILLS PARK



The first Sedge Wren documented in the park in 2020, which provides habitat for animal visitors
PHOTO: CAITLIN FIELD

IIT Studio:

Madison Weick

Anya Mehta-Rao

Caleb Hadley

Bowen Song

Chu-Fan Tzou

Huanyu Wang

Kevin Garcia

Galilea Ines

Jorge Mayorga

Jincheng Chen

Mansi Kishan Prajapati

Mohammad Arabmazar

Sadiqul Islam Shehab

Maurice Gaston

Ruth Muniz

Persefoni Stamatis

Zhicong Fang

Omar Rodriguez

IIT Teaching Team:

Dirk Denison

Maurice Cox

Maria A. Villalobos H.

Justin Degroff

Design

Los Pinos Design Group:

Persefoni Stamatīs

Jorge Mayorga

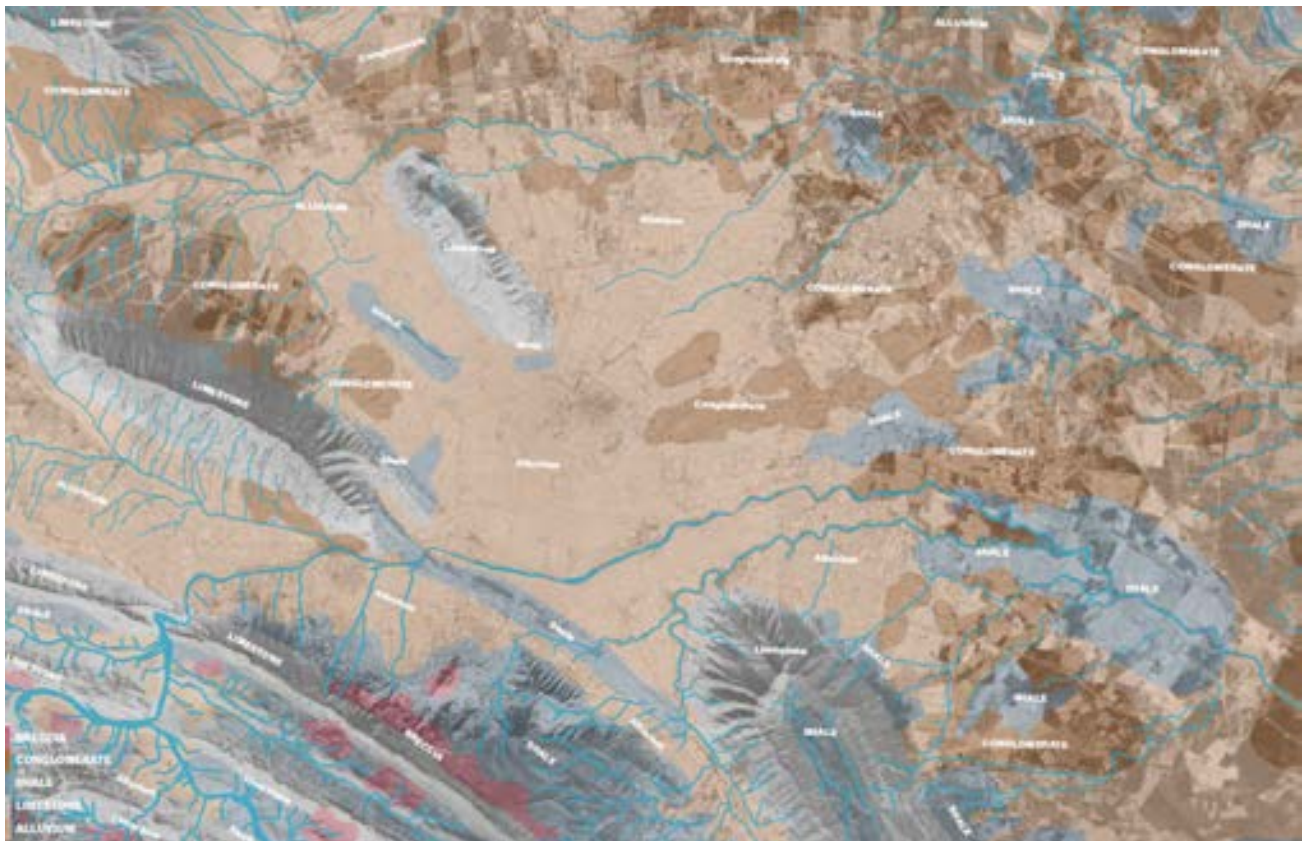
Chu-Fan Tzou

4.1. Diagnosis for Monterrey

To begin, my studio started analyzing and examining existing conditions for the neighborhoods in Monterrey, Nuevo Leon, through learning the history and current conditions of the city. We examined various aspects, including water pathways, historical development, agriculture, elevations, education, industries, vegetation, and soil composition. Our findings discovered that the essence of Monterrey cannot be identified with simple categories, it is made up by the contributions of the people who reside there.

A significant feature of Monterrey is the Arroyo Seco, also known as the Dry Creek, which is a cement-lined waterway that runs through the city. The creek lives up to its name since it is often filled with little to no water and is rather accumulating trash. The creek converges with Rio La Silla, a river running through the city.

Each group in our studio was then assigned a site along the Arroyo Seco in Monterrey with the intent of restoring not only the Dry Creek but also revitalizing the surrounding community.



Section of soil conditions in a mountain of Monterrey, Mexico (above), soil map of Monterrey, Mexico (below)



Location of Monterrey, Nuevo Leon, Mexico



Vegetation map of Monterrey, Mexico



Existing conditions of the Arroyo Seco

PHOTO: AGENCIA REFORMA



Volunteers gathering trash from the Arroyo Seco
PHOTO: LILIANA CAVAZOS



Los Pinos in La Campana
PHOTO: VALOR COMPARTIDO



4.2. Diagnosis for Campo los Pinos

Our group then proceeded to use the information we learned about Monterrey as a whole to focus on our specific site: Campo los Pinos, located between the neighborhoods Cerro de la Campana and Altamira. We analyzed the adjacent neighborhood Cerro de la Campana, paying close attention to the local, along with the site itself.

Campo los Pinos is a vacant lot located between these two neighborhoods, used with multiple purposes. Our group was fortunate enough to visit the site and engage with the local community, who shared valuable insights about the site's current uses and drawbacks. According to the residents, Campo los Pinos is currently used for political gatherings, markets, concerts, soccer, and primarily as an area to park. However, locals also expressed concerns about the safety of the site, regarding violence. This issue impacts the safety and accessibility of Los Pinos, potentially limiting the community.

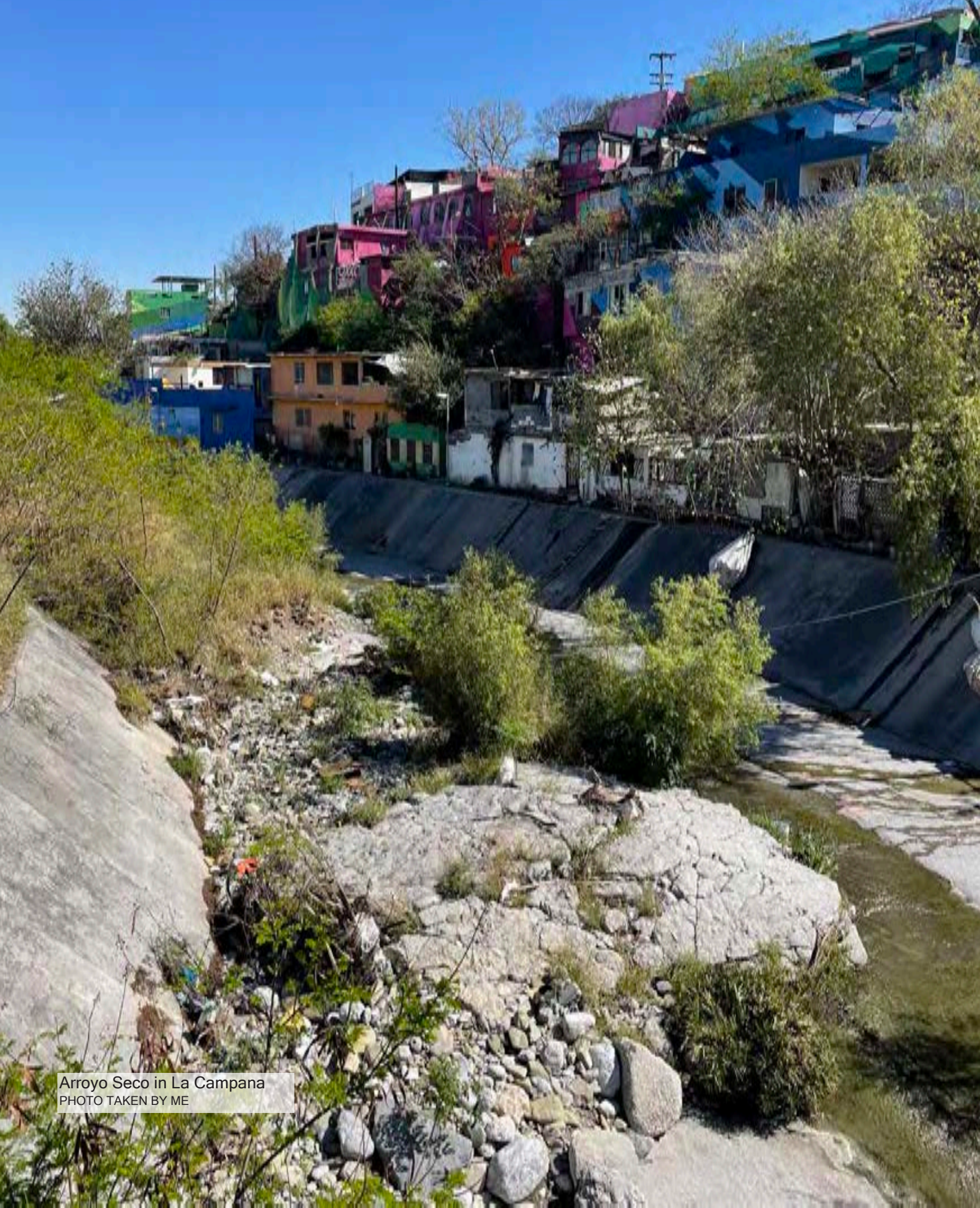
Campo los Pinos gathered its name, which literally translates to “the pines,” from the native pine trees that exist on the site. Historically, the site was an orange tree orchard with few pine trees. Understanding this history is significant as it provides an opportunity to reconnect the site with its roots through the reintroduction of native vegetation.

In addition to these factors, our conversations with the community revealed a strong desire for the lot to be transformed into a vibrant, safe, and more accessible area that can serve as a hub for social along with recreational activities.



La Campana, Monterrey, Mexico

Shown below is the Arroyo Seco which sits directly adjacent to Los Pinos. This creek is encased in concrete, disrupting the natural water flow and preventing the flourishing of nature. The concrete lining also enables flood risks, as it prevents water from being absorbed into the ground.



Arroyo Seco in La Campana
PHOTO TAKEN BY ME



Nogal de nuez lisa
Carya illinoensis



Mozquite
Prosopis laevigata



Montezuma cypress/ Ahuehuate o Sabino
Taxodium mucronatum



Tepoguaje
Leucaena pulverulenta



Espino
Acacia berlandieri



Carretilla
Medicago polymorpha



Higuera
Ricinus communis



Arundo
Arundo donax



Coquillo
Cyperus rotundus



Epa
Chenopodium

Campo los Pinos Existing Section Perspective

Depicting the existing conditions and vegetation of Campo los Pinos

ÁRBOLES Y ARBUSTOS EN CALLE TREES AND SHRUBS ON STREET



Hierba de donax
Donax



Hierba de Ambrosioides
Ambrosioides



Huizache/Espinillo blanco
Acacia farnesiana



Guamúchil
Pithecellobium dulce



Álamo de río
Platanus occidentalis mexicana



Mazquite
Prosopis laevigata



Bugambilia
Bougainvillea sp.



Ortiga
Urtica Urens



Escobilla
Fraxinus greggi



Jaboncillo
Sapindus saponaria

4.3. The Green Prescription for Campo los Pinos

Our design for Los Pinos provides the framework for improving the environment and enhancing the quality of life in the underinvested neighborhoods of Monterrey, Nuevo Leon, Mexico. To promote justice in the low-income informal neighborhoods of Cerro de la Campana and Altamira, our group worked with the people that inhabit the neighborhoods and plant landscapes that foster a thriving, self-sustainable, and integrated community.

Our design weaves together the people of Cerro de la Campana and Altamira through a green space by providing residents with native plants, trees, and fruit bearing trees. This approach reconnects the site with its history and enhances the environment while capturing and filtering runoff water, reducing the effects of summer heat, helping cleanse the polluted air, and providing fruits for the community. Preserving the existing trees while reintroducing the native oak and pine woodlands, mesquite woodland and desert scrubland will regenerate connections to nature and revive native ecosystems. These efforts will help bring back the native flora and create spaces where people can come together and feel a sense of community.

Our design involves carving into the site's elevation to reach the same level of the creek, along with an underlying bioswale and other pockets for water. This will improve water retention and expand park space as the adjacent creek is to be restored with greenery. A large multi-use space in the center along with carved seating on the sides will accommodate large gatherings and recreational activities. Also, an added bike path running along the creek will connect it to surrounding neighborhoods.

Additionally, our plan involves depaving the concrete environment of the creek to allow space for native vegetation to bloom. This project proposal will be an example of rewilding the concrete urban environment and creating a flourishing community that addresses climate change, creates a sense of belonging, encourages leadership, and cultivates a collective responsibility in caring for our planet, which is what The Green Prescription advocates for.



Campo los Pinos Currently
PHOTO: CHUFAN TZOU



Campo los Pinos Design Render



Campo los Pinos Existing Plan

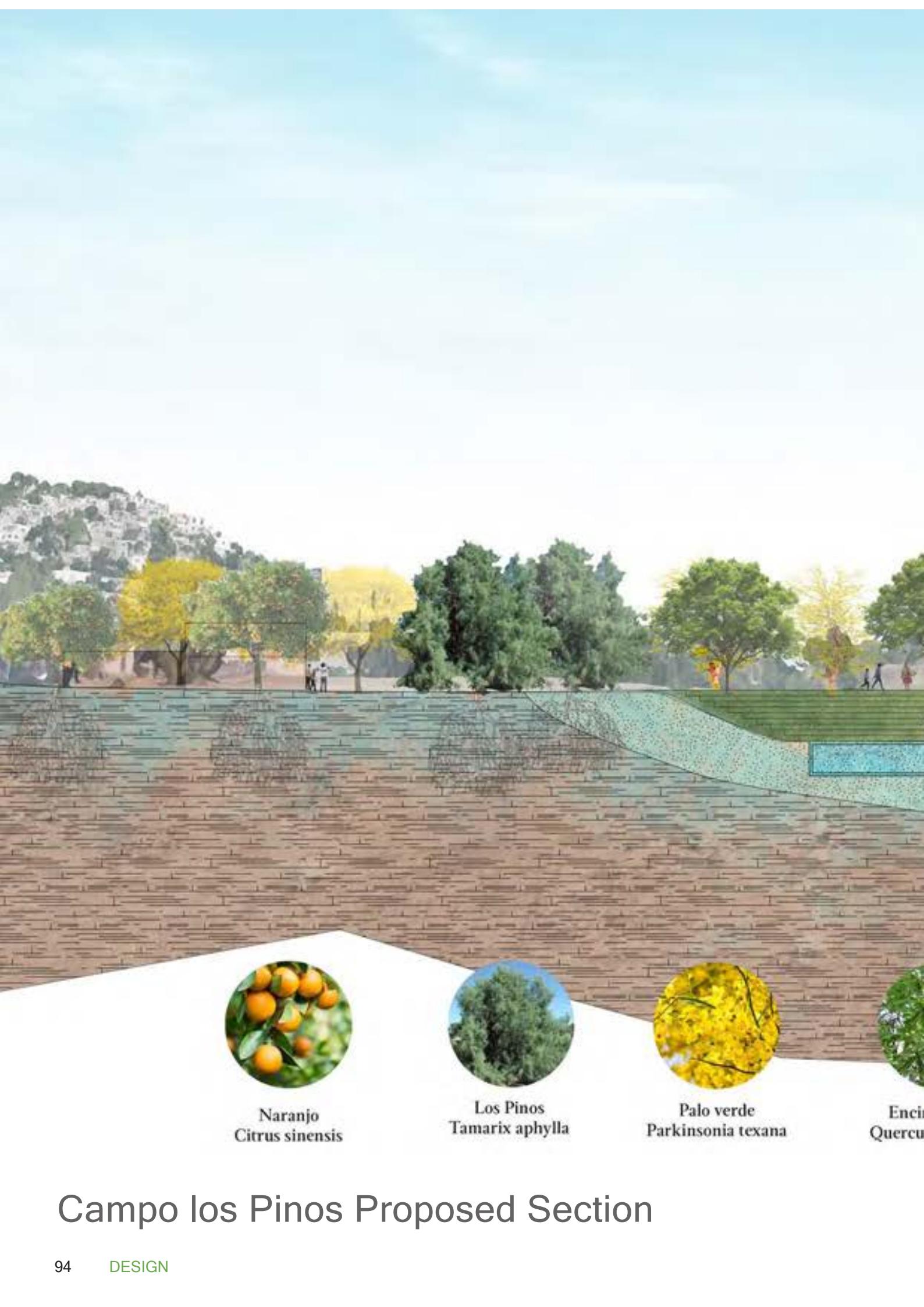


SATELLITE IMAGE FROM GOOGLE EARTH



Campo los Pinos Proposed Plan





Naranjo
Citrus sinensis



Los Pinos
Tamarix aphylla



Palo verde
Parkinsonia texana



Encino
Quercus

Campo los Pinos Proposed Section



Palo de asta
Lysophylla

Pino blanco
Pinus pseudostrobus

Huizache/Espinillo blanco
Acacia farnesiana

Corona de San Pedro
Cornus florida

5. Conclusion

The Green Prescription acknowledges the benefits that green spaces have on the environment and on human health. For humans, these spaces improve overall mental, physical, and sociability, while aiding in healing. Environmentally, green spaces contribute to soil conservation, water management, and biodiversity preservation. Similarly for both humans and the environment, green spaces help sequester carbon, leading to heat relief, better air quality, and acting as buffers for noise pollution.

Building on Charles Waldheim's concept of integrating green spaces into urban design, The Green Prescription takes this further by turning vacant land into functional green areas. The Green Prescription has goals of enhancing public health, increasing community interactions, mitigating environmental impacts, and promoting biodiversity. This idea has been shown to have positive impacts for mental health and the environment in the city of Philadelphia. The Green Prescription uses the process of site-specific differential diagnosis, adapting to the needs of each location.

The Green Prescription was then applied to Campo los Pinos in Monterrey, Nuevo Leon, Mexico, with my studio group. This design called for the preservation of existing trees while reintroducing native orange trees, native plants, enhancing water management, and creating spaces for community gatherings and natural exploration. This project displays the practical and beneficial outcomes of The Green Prescription in the real-world setting.

The Green Prescription displays the role that green spaces have in enhancing the urban environment, emphasizing the importance of integrating nature into the built environment to benefit both humanity and the planet.

Sources

American Planning Association. (n.d.). Combating climate change through vacant lot restoration. Retrieved from <https://planning.org/blog/9218357/combating-climate-change-through-vacant-lot-restoration/>

American Society of Landscape Architects. (2018). Paris Parks: A Laboratory of Urbanism and Ecology. Landscape Architecture Magazine. Retrieved from https://www.asla.org/uploadedFiles/CMS/About__Us/Diversity/LAM_04Apr2018_F3-ParisParks-spreads.pdf

Beft21 City Parks. (2021, December 7). City Parks and Their Impact on the Environment | Don't Waste My Energy!. Retrieved from dontwastemy.energy/2021/01/20/city-parks-and-their-impact-on-the-environment

Cleveland Clinic. (2023, January 10). Differential diagnosis. <https://my.clevelandclinic.org/health/diagnostics/22327-differential-diagnosis>

Dee, R. (n.d.). 6.4 Buffers for Noise Control. U.S. Forest Service. Retrieved from www.fs.usda.gov/nac/buffers/guidelines/6_aesthetics/4.html

Fo. (n.d.). Field Operations - project_details. <https://www.fieldoperations.net/project-details/project/freshkills-park.html>

Food and Agriculture Organization of the United Nations (FAO). (n.d.). Soils store and

filter water - Improving food security and our resilience to floods and droughts. RTP-RNE-WS. Retrieved from www.fao.org/platforms/water-scarcity/Knowledge/knowledge-products/detail/soils-storeand-filter-water---improving-food-security-and-our-resilience-to-floods-and-droughts/en

Freshkills Park. (n.d.). The park plan. <https://freshkillspark.org/the-park/the-park-plan>

Freshkills Park Alliance. (n.d.). Chronology of the Freshkills Park Site. Retrieved from <https://freshkillspark.org/the-park/chronology-of-the-freshkills-park-site>

International Union for Conservation of Nature (IUCN). (2023). Embracing Biodiversity: Paving the Way for Nature-inclusive Cities. Retrieved from iucn.org/story/202305/embracing-biodiversity-paving-way-nature-inclusive-cities

Landezine. (n.d.). Martin Luther King Park by Atelier Jacqueline Osty & Associés. Landezine. Retrieved from <https://landezine.com/martin-luther-king-park-by-atelier-jacqueline-osty-associes/>

National Grid. (n.d.). Energy Explained. Retrieved from www.nationalgrid.com/stories/energy-explained/what-carbon-sequestration

Paris & Métropole Aménagement. (n.d.). Le parc Martin Luther King. Retrieved from <https://archive-clichy-batignolles.paris-et-metropole-amenagement.fr/le-parc-martin-luther-king.html>

Paudel, S., & States, S. L. (2023). Urban green spaces and sustainability: Exploring the ecosystem services and disservices of grassy lawns versus floral meadows. *Urban Forestry & Urban Greening*, 84, 127932. <https://doi.org/10.1016/j.ufug.2023.127932>

Public Space. (n.d.). Clichy-Batignolles - Martin Luther King Park. <https://www.publicspace.org/works/-/project/m102-clichy-batignolles-martin-luther-king-park#>

Räsänen, J., et al. (2013). Effects of species-specific leaf characteristics and reduced water availability on fine particle capture efficiency of trees. *Environmental Pollution*, 183, 64-70. <https://doi.org/10.1016/j.envpol.2013.05.015>

Soil Science Society of America. (n.d.). Green Infrastructure. Retrieved from www.soils.org/about-soils/green-infrastructure

The High Line. (n.d.). History. Retrieved from <https://www.thehighline.org/history/>

The Rapidian. (n.d.). Why parks matter: How our parks affect city life. The Rapidian. Retrieved from www.therapidian.org/why-parks-matter-how-our-parks-affect-city-life

Time Magazine. (n.d.). Urban greening and depression study. Retrieved from <https://time.com/5341975/urban-greening-depression-study/>

U.S. Department of Energy. (n.d.). DOE Explains...the Carbon Cycle. [Energy.gov](http://energy.gov). Retrieved from www.energy.gov/science/doe-explainsthe-carbon-cycle#:~:tex-

[t=The%20carbon%20cycle%20is%20the,-makes%20life%20on%20Earth%20possible](#)

Van Den Berk Boomkwekerijen. (2023, August 7). How trees reduce noise pollution in urban areas. Retrieved from www.vdberk.nl/inspiratie/geluidsvervuiling-verminderen-met-bomen/?trk=article-ssr-frontend-pulse_x-social-details_comments-action_comment-text

Waldheim, C. (2016). *Landscape as Urbanism: A General Theory*. Princeton University Press.

